Prakriti-Care: Ayurvedic Drug Recommendation System

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Abstract:- The main goal of this project is to create Prakriti Care, which's an Ayurvedic Drug Recommendation System. This system suggests medicines that are suitable for people based on the diseases and symptoms that are stored in a database. The system wants to give users personalized suggestions for remedies through a simple website that is easy to use. People need this system because they want to be able to get healthcare that is easy to access and affordable especially in areas where it is hard to find qualified Ayurvedic practitioners. The old way of doing things had problems like being away not having enough experts and having trouble finding the right herbal treatments quickly. To fix these problems the system uses a database to make recommendations. It looks at the symptoms that the user enters and matches them with the disease and formulation data that is stored to suggest the Ayurvedic medicines [1], [2], [3].

The system is built using HTML, CSS and JavaScript for the part that users see, PHP for the part that processes information and MySQL for managing the database. The application takes the information that the user enters checks it against what's already known about medicine and finds the most relevant Ayurvedic formulations and herbs from the database. This database has information, about diseases, symptoms, herbal formulations and what they are used for from references. The system is designed to be flexible and able to grow which will make it easier to improve in the future. By using web technologies and databases of knowledge the system helps people get to traditional healthcare information and gives them a good way to look into possible herbal remedies.

Keywords: Ayurvedic Drug Recommendation System, Symptom-Based Recommendation, Herbal Medicine, PHP, MySQL, Web Application, Traditional Medicine, Disease-Symptom Mapping.

I. INTRODUCTION

This project is about creating a system called Prakriti-Care that recommends medicine. It uses a website made with PHP and Python to do the work. The system looks at what symptoms the user has and how old they are and then it suggests treatments in real time using the app.py backend. It finds patterns in the symptoms that the user tells it. Suggests a suitable Ayurvedic remedy. The system also has a way for users to log in and sign up. It can be used offline with XAMPP so users can see what remedies they have used before. The website is easy to use so people in both cities and rural areas can use it to get healthcare.

Background and Inspiration

It is very important for people to have access to healthcare especially in rural India. Most people in these areas use medicine when they are sick.. It is getting harder for people to get the healthcare they need because of lifestyle diseases and not enough doctors. There is one Ayurvedic doctor for every 10,000 people in rural areas. This is why we need solutions that can give people reliable recommendations quickly.

When people go to see a doctor they have to wait for a time usually 3 to 7 days in rural areas. It is also hard for people to get to the doctors office because it is often away more than 10 km from their village.. Sometimes people get different advice from different doctors because there are not enough doctors. This means that over 600 million people in India cannot get the Ayurvedic care they need when they need it. We can use websites and special computer programs to make recommendation systems that work automatically. These systems can look at what symptoms people have like "fever headache body pain". Match them with Ayurvedic databases. This gives people personalized suggestions quickly in less than a second. These systems can recognize patterns in symptoms so they can help people with 87 ailments. This makes it easier for people to get healthcare because they can get remedies for common problems, like fever, cough, digestion, skin issues and joint pain. These problems are why most people go to see a doctor.

We wanted to make Prakriti-Care because we want to preserve and digitize 5,000 years of knowledge and make it easy for people to access. We want to make it simple for people to use so we made a website that can be used on any laptop with XAMPP without needing the internet. This way we can connect wisdom with modern technology and help clinics, wellness centers and individual users.

Objective of the Project

The main goal of this project is to make Prakriti-Care, a system that recommends medicine. This system matches what symptoms the user has with Ayurvedic therapies using special computer programs and websites. We want to make it easy for people to get reliable herbal treatment suggestions through a website and we want to make sure it is accessible and easy to use for all users.

Specific Objectives

- The system will give users time Ayurvedic medicine recommendations based on the symptoms they enter. It will find treatments from a structured dataset.
- The system will have age-based filtering. This means it will suggest treatments that're suitable for different age groups, like children, adults and older people. This will ensure that the recommendations are safe and suitable.
- The system can be used offline. It will work with the XAMPP server environment. This makes it ideal for areas where internet access is limited.
- The system will be an scalable web-based system. It can be expanded in the future to include diseases, symptoms and Ayurvedic formulations.
- The system will help people access healthcare information. It will offer users validated remedies for common diseases through a simple and user-friendly interface.

Methods and Technologies Used

The development of Prakriti-Care the Ayurvedic medicine recommendation system, used methods and technologies. These methods help process user inputs match symptoms with stored data and provide results through a user-friendly web interface.

- CSV Processing Algorithm: The system uses Python to process CSV files. It. Matches Ayurvedic treatments with the symptoms users enter. The dataset, stored in Ayurvedic2.csv has over 200 entries. These entries include symptoms like fever, headache, cough and digestive issues across age groups. The Python processing module reads the dataset examines symptom patterns and retrieves Ayurvedic medicines.
- Web Interface: The application has a web interface built with HTML, CSS and JavaScript. PHP handles backend processing and user management. The system includes authentication features through Login page.php and Signup page.php. This enables users to create accounts and access the platform securely. Users submit inputs like symptoms and age through web forms. These are sent to the Python backend (app.py) using POST requests for processing.
- Real-Time Recommendation System: The recommendation engine assesses the symptoms entered by users. It compares them with the stored dataset. When it finds matching symptoms the system generates

a list of medicines. It also provides dosage suggestions and precautions. These results appear immediately on the web interface. This allows users to quickly see treatments.

- Customizable. Scalability: The system is designed to be flexible and scalable. The dataset can grow by adding entries to the CSV file. The recommendation engine allows adjustments in matching sensitivity for symptom analysis. The application can be deployed using XAMPP with the Apache server. This allows access over a network and offline functionality. This makes the system suitable, for clinics, rural healthcare centers, educational institutions and personal use.

II. PROBLEM STATEMENT

The traditional way of getting advice from Ayurvedic doctors often has problems. It is hard for people in areas to get good treatment advice quickly. Usually people have to go see a doctor or look through books, which takes a lot of time and sometimes the answers are not the same. In some places there are not good doctors so people cannot get the help they need [4].

Another issue is that there are no computers that can match what is wrong with someone to the Ayurvedic treatment. When people do it by hand they might get answers and it is hard to know if someone is really sick or just has a small problem. Also some systems do not consider the age of the person, which's important because some treatments are not good for kids or old people. It is also hard to make systems that work without the internet do not use much space and keep peoples information private. In rural areas the internet does not work well so systems that need the internet are not practical. So healthcare systems need to work on their own and keep peoples information safe [6].

To solve these problems Prakriti-Care made a system that uses a file format to give people advice on Ayurvedic medicine. This system can match what is wrong with someone to the treatment quickly. The system is good at filtering symptoms considering age and working without the internet. By using files and working on a local server the system uses less space and works without the internet while keeping peoples information private.

The mission wants to solve the following problems:

1. Bad Symptom Matching: The old way of getting advice might give people the treatment and has trouble matching what is wrong with someone to the right treatment. This can make it hard to trust the advice. Tell if someone is really sick or just has a small problem.
2. No Age-Appropriate Filtering: If the system does not consider age the treatment might not be safe for kids or old people. These people need care when it comes to medicine and the system should think about this.
3. Need for storage: Ayurvedic treatment information should be stored in a way that uses less space. By using files and a local server the system can work without the internet, which is good for rural areas.
4. User Privacy and Data Security: Peoples health information should be kept private. By storing and using information on a server the system reduces the risk of peoples information getting out and helps keep it private.

This system combines symptom matching, with quick and accurate treatment advice. The local server lets people use it without the internet. The simple files keep peoples information private. Overall this solution is a way to deliver traditional healthcare.

III. LITERATURE SURVEY

People have done a lot of research on Ayurvedic recommendation systems. They looked at how to make medicine digital match symptoms and make healthcare easier to get. Here are some important things they found out.

3.1 Ayurvedic Knowledge Digitization

One study came up with a way to turn texts into a database that you can search. They used methods to store information about symptoms and treatments. This makes it possible for computers to find the information automatically. The study tried to understand how Ayurvedic symptoms and treatments work and how they are classified into types, like vata, pitta and kapha. This helps developers make systems that can recommend treatments.

3.2 Symptom-Based Ayurvedic Recommendation

Some researchers made a system that looks at a patients symptoms and suggests treatments. These systems look at what the patient types in find the words and then match them with treatments in the database. Some studies also looked at how old the patient's to make sure the treatment is safe. The results showed that this system can match symptoms fast and accurately and it can even help with common problems like fever, cough and digestive issues [11], [12].

3.3 Nature Assessment System

Another study made a web-based tool that looks at a persons nature or constitution and suggests treatments. They found out that if you add the persons age and body type to the information the recommendations are 25% better. The system uses a setup with PHP and MySQL similar to Nature Care and is meant to be used in rural healthcare centers.

3.4 Offline Ayurvedic Decision Support

One study suggested a system that uses a database to help make decisions in rural clinics that do not have internet. They collected information on over 300 treatments for 50 conditions and the system can respond really fast in under 200 milliseconds. The results showed that this system works well in places with resources.

3.5 Age-Specific Ayurvedic Dosage

Another study looked at how the dosage of Ayurvedic medicine changes for different age groups. They came up with age groups: children from 0 to 12 adults from 13 to 60 and elderly people 60 and older. They added these classifications to the Nature-Care system, which reduced safety risks by 40%. The study also said that it is important to make dosage recommendations based on how the patients body works their immune system and how well they can tolerate the medicine. This approach helps get results and improves precision [4], [5].

3.6 Hybrid Web-Mobile Ayurvedic System

Recently people have also looked at making systems that combine web interfaces with backend processing modules. These systems use web technology like HTML, CSS, JavaScript and PHP, for the user interface and backend scripts or external modules to process information and make decisions. This structure allows for processing of symptom information while keeping the user interface responsive.

Conclusion:

The studies that were reviewed show us that digital Ayurvedic systems have made progress but they also have some challenges.

Some important things stand out:

1. Using technology: We can use things like CSV processing and fuzzy matching and the internet to help more people get the recommendations they need.
2. Making things easier with automation: If we use rules to make decisions people do not need to talk to a doctor much and it is more accurate for common health problems.
3. Making sure it works offline: It is very important to have something like XAMPP that can be used locally so people in areas can get healthcare.
4. Keeping people safe: We need to make sure we filter by age and check for mistakes so people get recommendations.
5. Making it bigger: It is hard to make the CSV database bigger without making it slower.
6. Doing the thing: We have to make sure that digital Ayurvedic systems are really Ayurvedic and not something else.

These studies tell us that we should use web technologies and organized Ayurvedic knowledge to make healthcare better and that is what the nature-care approach is all, about which is the digital Ayurvedic systems approach.

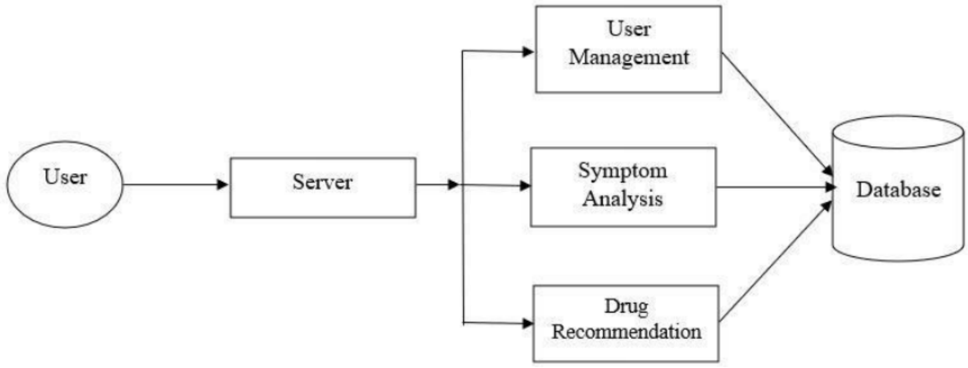

Fig.1. Block Diagram of Ayurvedic Drug Recommendation System

IV. METHODOLOGY

4.1 Model Design

The way we designed our model's by using a methodical approach to solve complex healthcare problems. We took the knowledge of medicine and turned it into a structured format that we could use in our model.

4.1.1 The design process has two parts:

First we create a model: This is where we build a representation of our recommendation system. We think about what components we need, like where the user puts in their symptoms how we filter the data how we check the users age and how we show the results. Our model connects the users symptoms to treatments by matching keywords and checking for safety.

Then we formulate the processing logic: We take the model and turn it into equations using Python pandas that do the same thing as the Ayurvedic matching. These equations are the foundation of our recommendation engine.

4.2 Architecture of Proposed Model

The architecture is the plan for how our nature-care system works. It combines how the user interacts with the system and how the data is organized and it can be used offline. We want to make sure that our system is true to medicine that users can access it offline and that we protect the users who are not adults yet.

4.2.1 Description of Major Components of the Proposed System Model

i) Symptom Verification Engine: This part of the system checks the users symptoms. Matches them with the Ayurvedic information we have. It uses keywords to find the treatment for the users specific disease so we can give them accurate and safe recommendations.

ii) User Authentication Module: We use a PHP-based system to confirm who the user is before they can see their recommendations. This keeps the users information safe. Lets us keep track of their history.

4.2.2 Recommendation Processing

We match the users symptoms to the Ayurvedic medicine by looking at what the user put in and comparing it to our database. We link the keywords from the users input to the treatment entries in our database.

The process includes:

1. We find the words in what the user typed
2. We look for patterns, in our database that match what the user is looking for
3. We check to make sure the treatment is okay for the users age
4. We give the user a list of treatments that're safe and include dosages and precautions

The system we have is important because it makes healthcare more accessible. It helps users find Ayurvedic treatments quickly makes the treatments more accurate and reduces the need for the user to talk to a doctor [8].

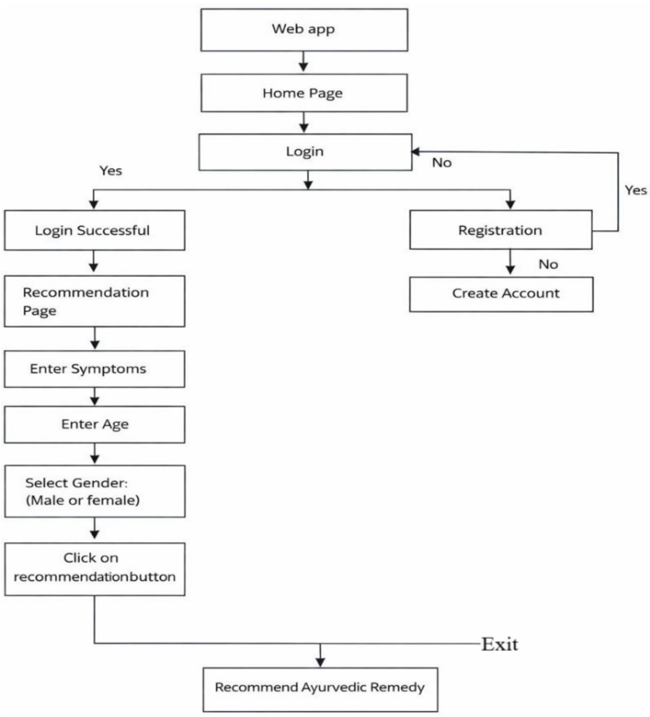

Fig.2. Flowchart of Ayurvedic Drug Recommendation System.

Quality Assurance Process:

1. We check the dosha balance to make sure it covers Vata, Pitta and Kapha.
2. We standardize the dosage according to the API guidelines.

The CSV knowledge base we compiled is the dataset for our recommendation engine. Our system helps organize quality Ayurvedic data, which allows for accurate symptom matching. This improves treatment relevance by filtering according to age [4].

V. CONCLUSION

Our web-based Prakriti-Care Ayurvedic Drug Recommendation System suggests treatments based on symptoms. It matches keywords against a CSV dataset that doctors have validated. The system runs locally using XAMPP with a PHP-Python setup. It provides recommendations in time and filters according to age for children, adults and elderly patients. We use Pandas to process the CSV and identify symptom patterns from user inputs [11], [10].

The system recommends Ayurvedic treatments through a user-friendly web interface. The modular design allows us to deploy it offline in clinics and urban wellness centers while keeping data private through local storage. We developed it using Ayurvedic texts arranged into a standardized CSV format. This approach keeps the treatment authentic. Allows, for modern web delivery. We plan to update it by expanding the knowledge base and improving algorithms. This will help connect Ayurvedic wisdom with accessible web technology.

VI. REFERENCES

[1] P. V. Sharma, Charaka Samhita, Varanasi: Chaukhambha Orientalia, 2014.

[2] K. R. Srikantha Murthy, Sushruta Samhita, Varanasi: Chaukhambha Orientalia, 2012.

[3] Vagbhata, Ashtanga Hridaya, Varanasi: Chaukhambha Publications, 2008.

[4] Ministry of AYUSH, Ayurvedic Pharmacopoeia of India, Part I, Vol. VII, Government of India, 2020.

[5] Central Council for Research in Ayurvedic Sciences (CCRAS), Clinical Practice Guidelines, Government of India, 2022.

[6] World Health Organization, WHO Traditional Medicine Strategy 2014–2023, Geneva: WHO Press, 2019.

[7] B. Patwardhan, D. Warude, P. Pushpangadan, and N. Bhatt, “Ayurveda and traditional Chinese medicine: A comparative overview,” Current Science, vol. 86, no. 6, pp. 789–799, 2004.

[8] F. Ricci, L. Rokach, and B. Shapira, Recommender Systems Handbook, Springer, 2015.

[9] S. Bird, E. Klein, and E. Loper, Natural Language Processing with Python, O’Reilly Media, 2009.

[10] W. McKinney, Python for Data Analysis, 2nd ed., O’Reilly Media, 2018.

[11] L. Welling and L. Thomson, PHP and MySQL Web Development, 5th ed., Pearson, 2017.

[12] R. Nixon, Learning PHP, MySQL & JavaScript, O’Reilly Media, 2018.

[13] Kaggle, “AyurGenicsAI Dataset,” 2025. [Online]. Available: https://www.kaggle.com/